

## PCT/PCM5: Pick the Right Filter

### 4: Interpret the Figure

The best result for salt-and-pepper noise is the median-filtered image, while the mean-filtered result is among the worst. The median filter is robust to extreme outliers such as the 0 and 255 values produced by salt-and-pepper noise, whereas the mean filter includes these outliers in its average and therefore smears their effect.

### 5: Mean Filter Calculation

A 3×3 mean filter averages 9 independent noise samples.

$$\sigma_{\text{out}} = \sigma / \sqrt{N} = 20 / \sqrt{9} = 20 / 3 \approx 6.7$$

Therefore, the output noise standard deviation is 6.7. The same averaging operation mixes pixels from both sides of a 1-pixel-wide edge, so the sharp edge becomes blurred/smoothed.

### 6: Prediction

Predicted ranking for salt-and-pepper noise:

Median > Gaussian > Mean

The median should rank first because it is resistant to the extreme 0/255 outliers produced by salt-and-pepper noise. The mean is strongly affected by these outliers because it directly averages them with neighboring pixels.